Abstract Algebra

Summary note of Dummit & Foote's Abstract Algebra

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Contents

Chapter 1

Introduction to Groups

1.1 Groups

Binary operation

1. A *binary operation* $\star$ on a set G is a map $\star : G \times G \to G$. We denote $\forall a, b \in G, \star(a, b)$ by $a \star b$.

2. A binary operation $\star$ on a set G is *associative* if

$$\forall a, b, c \in G, a \star (b \star c) = (a \star b) \star c$$

3. A binary operation $\star$ on a set G is *commutative* if

$$\forall a, b \in G, a \star b = b \star a$$

4. For a set G, $H \subseteq G$ and a binary operation $\star$ on G, if the restriction of $\star$ to H is a binary operation on H, H is said to be *closed* under $\star$.

Group

1. A *group* is an ordered pair $(G, \star)$ where G is a set and $\star$ is a binary operation on G, which satisfies

   (a) $\star$ is associative.

   (b) $\exists\ e \in G, \forall a \in G, a = a \star e = e \star a$

   (c) For e of (b), $\forall a \in G, \exists a^{-1} \in G, a \star a^{-1} = a^{-1} \star a = e$

2. Group $(G, \star)$ is called *abelian* when $\star$ is also commutative.

Direct product

Let $(A, \star)$ and $(B, \diamond)$ are groups. We can form group $A \times B$ called *direct product* of A and B by

$$A \times B = \{(a, b) | a \in A, b \in B\}$$

with following binary operation $\cdot$:

$$(a_1, b_1) \cdot (a_2, b_2) = (a_1 \star a_2, b_1 \diamond b_2)$$

Here, one can claim that $(A \times B, \cdot)$ is a group. The associativity of $\cdot$ is clear. And let's denote the identity of $(A, \star), (B, \diamond)$ by e_1, e_2 respectively. We can easily check that (e_1, e_2) is identity of $(A \times B, \cdot)$ and $(e_1, e_2) \in A \times B$. This holds group axiom (2). Finally, $\forall a \in A, b \in B$, we can compose (a^{-1}, b^{-1}). Then by the definition of $\cdot$, $(a, b) \cdot (a^{-1}, b^{-1}) = (a^{-1}, b^{-1}) = (e_1, e_2)$ holds. So, $(a, b)^{-1} = (a^{-1}, b^{-1})$, holds group axiom (3).

Propositions on Groups

Let $(G, \star)$ be any group. Following always holds.

1. The identity of G is unique.
2. $\forall a \in G$, a^{-1} is uniquely determined.
3. $\forall a \in G, (a^{-1})^{-1} = a$
4. $\forall a, b \in G, (a \star b)^{-1} = b^{-1} \star a^{-1}$
5. $\forall a_1, \cdots, a_n \in G$ the value of $a_1 \star a_2 \star \cdots \star a_n$ is independent of how the expression is bracketed.

All the proofs are easy.

Cancellation Laws hold in G

1. $au = av \implies u = v$
2. $ub = vb \implies u = v$

Group order

For any group G and $x \in G$, let's say the *order* of x (denote by $|x|$) by

$$|x| = \underset{n \in \mathbb{N}}{\operatorname{argmin}} \{x^n = 1\}$$

If such n does not exists, $|x| = \infty$.

1.2 Dihedral Groups

The definition of *dihedral group* is quite abstracted.

Dihedral Groups

One natural, and important structure is symmetries of geometric objects. For $n \in \mathbb{Z}^+, n \geq 3$, the *dihedral group* D_{2n} be the set of symmetries on regular n-gon, where a symmetry is any rigid motion.

Now let's imagine that we label each vertices of such n-gon as $\{1, 2, \cdots, n\}$. Then, each symmetries can be viewed a permutation σ on $\{1, 2, \cdots, n\}$. Then now we can formulate the group D_{2n} with composition $\circ$. Let $s, t \in D_{2n}$ be two symmetries. Composition $s \circ t = st$, is defined by apply t and then s.

One interesting thing is that, $|D_{2n}| = 2n$. It can be proved. In every symmetry, vertex 1 and 2 are adjacent. So, our job is clear. Choose the position of vertex 1. (n cases) And then, choose the position of vertex 2 which adjacent with vertex 1. (2 cases) So, we have $2n$ cases total, implies that $|D_{2n}| = 2n$. Or, we can enumerate what such $2n$ symmetries are. n symmetries are came from rotation by $2\pi i/n, i \in \{0, \cdots, n-1\}$. Remaining n symmetries are came from flip(reflection), with n axis of n-gon.

From now on, we'll fix our setting to understand D_{2n} further. Fix a regular n-gon centered at the origin on $\mathbb{R}^2$ and label the vertices in clockwise manner, from 1 to n. And denote $r \in D_{2n}$ be a $\frac{2\pi}{n}$ rotation. And denote $s \in D_{2n}$ be a reflection via axis containing vertex 1. We can observe following propositions.

Propositions on D_{2n}

1. $1, r, r^2, \cdots, r^{n-1}$ are distinct, and $|r| = n$
2. $|s| = 2$
3. $s \neq r^i$ for any $i \in \{0, 1, \cdots, n-1\}$
4. $sr^i \neq sr^j$ for all $0 \leq i, j \leq n-1$ with $i \neq j$. Consequence of this is that,
$$D_{2n} = \{1, r, r^2, \cdots, r^{n-1}, s, sr, sr^2, \cdots, sr^{n-1}\}$$
5. $rs = sr^{-1}$, So D_{2n} is *non-abelian*
6. $r^i s = sr^{-i}, 0 \leq i \leq n$

Everythings are not hard to prove.

Now, let's discussion about *generators* and *relations*. As we can see in above study, to represent all elements of D_{2n}, two element r, s are sufficient. Similarly, we introduce the notions of *generators* and *relations* for arbitrary groups. Although we recall both notions later, let's bring the notions of this basis-like things.

Generator of G

For a group G, $S \subset G$ is called a set of *generators* if every element of G can be written as a (finite) product of elements of S and their inverses. Also we write $G = \langle S \rangle$ and say that G is *generated by* S.

For clear understanding, $\mathbb{Z} = \langle 1 \rangle, D_{2n} = \langle r, s \rangle$.

Relation in G

Any equations in group G that the generators of G satisfy are called *relations* in G.

For example, we have 3 relations in D_{2n} :

$$r^n = 1, s^2 = 1, rs = sr^{-1}$$

And any other relation between elements of D_{2n} can be derived from these three. (later) By the way, this notion enable us to represent group in simple way. (Only few elements and equations) This leads us to following definition.

Presentation of G

Let's imagine that a group G is generated by $S \subseteq G$ and there are some collection of relations $R_1, \cdots, R_m$ such that any relation among the S is consequence of them. In this case, we write

$$G = \langle S | R_1, R_2, \cdots, R_m \rangle$$

and call this by *presentation* of G.

In this step, this notion is quite verbose. So, we'll discuss it later of this book.

1.3 Symmetric Groups

Now let's move on to next step. Actually our goal is abstraction of natural notions. In this section, we'll derive set of all bijections(or permutations).

Symmetric Group on Ω

Let Ω be any set. Then define S_Ω by the set of all bijections from Ω to Ω. We can also understand S_Ω is set of all permutations of Ω. For the binary operation $\circ$ which is function composition, $(S_\Omega, \circ)$ is a group. (Check!) We call $(S_\Omega, \circ)$ by *symmetric group* on Ω.

One natural consideration is that, when $\Omega = \{1, 2, \cdots, n\}$. In this case, we denote the symmetric group by $(S_n, \circ)$ and call it *symmetric group of degree n*. We already know that $|S_n| = n!$

Let's move on. Can we find some interesting observations from each $\sigma \in S_n$?

Cycle decomposition

A *cycle* is a string of integers represents the element of S_n which cyclically permutes these integers and fixed all others. For example, a cycle $(a_1, \cdots, a_m) \in S_n$ is the permutation such that sends $a_1 \mapsto a_2, \cdots, a_{m-1} \mapsto a_m, a_m \mapsto a_1$
In general, we can decompose for each $\sigma \in S_n$ into k cycles, means that

$$\forall \sigma \in S_n, \sigma = (a_1\ a_2 \cdots a_{m_1})(a_{m_1+1} \cdots a_{m_2}) \cdots (a_{m_{k-1}+1} \cdots a_{m_k})$$

With this, we can easily compute $\sigma(x)$ for any $x \in [n]$. Find the cycle that x is contained, then the immediately right-next element is $\sigma(x)$. This decomposition is called *cycle decomposition of σ*

We can easily obtain the cycle decompositioning algorithm, so skip. And with this, we can easily compute cycle decomposition of σ^{-1}. Just write the numbers in each cycle of the cycle decomposition of σ in reverse order. i.e.,

$$\sigma^{-1} = (a_{m_1}\ a_{m_1-1} \cdots, a_1)(a_{m_2}\ \cdots a_{m_1+1}) \cdots (a_{m_k} \cdots a_{m_{k-1}+1})$$

Also, we can observe that $\forall n \geq 3, S_n$ is not abelian group. This is because that $(1\ 3) \circ (1\ 2) \neq (1\ 2) \circ (1\ 3)$. And note that, we can view σ by composition of each cycles. (since cycles are permutations of S_n itself) Moreover, they are independent to order. i.e., *disjoint cycles commute*. And also, each cycles are independent to cyclic permutation of inner elements. i.e., $(1\ 2\ 3) = (2\ 3\ 1) = (3\ 1\ 2)$

Exercise 10

Prove that if σ is the m-cycle $(a_1, a_2, \cdots, a_m)$, then for all $i \in \{1, 2, \cdots m\}, \sigma^i(a_k) = a_{k+i}$, where $k+i$ replaced by its least positive residue mod m. Deduce that $|\sigma| = m$.

Proof. Let's denote the least positive residue mod m by $R_m^+ : \mathbb{Z} \to [m]$. Note that, $R_m^+(m) = m$, and $R_m^+(m+1) = 1$. Then we can immediately obtain that $R_m^+(x+m) = R_m^+(x)$. Then, let's prove by mathematical induction w.r.t. i. For $i = 1$,

$$\sigma(a_k) = \begin{cases} a_{k+1} & 1 \le k < m \\ a_1 & k = m \end{cases}$$

However,

$$R_m^+(k+1) = \begin{cases} k+1 & 1 \le k < m \\ 1 & k = m \end{cases}$$

by definition. So, $\sigma(a_k) = a_{R_m^+(k+1)}$ so the base case holds. Now, suppose that

$$\sigma^i(a_k) = a_{R_m^+(k+i)}$$

Then,

$$\sigma^{i+1}(a_k) = \sigma(\sigma^i(a_k)) = \sigma(a_{R_m^+(k+i)}) = \begin{cases} a_{R_m^+(k+i)+1} & 1 \le R_m^+(k+i) < m \\ a_1 & R_m^+(k+i) = m \end{cases}$$

However, for $1 \le R_m^+(k+i) < m$, $R_m^+(k+i)+1 = R_m^+(k+i+1)$ and $R_m^+(k+i) = m$, then $1 = R_m^+(k+i+1)$. So we can write that

$$\sigma^{i+1}(a_k) = a_{R_m^+(k+i+1)}$$

This completes the proof. □

Exercise 11

Let σ be the m-cycle $(1, 2, \cdots, m)$. Show that σ^i is also an m-cycle if and only if i is relatively prime to m.

Proof. We can imagin σ as $a_k = k$ in Exercise 10. Then here, we can obatain that

$$\sigma^i(k) = R_m^+(k+i)$$

So that, we can construct cycle decomposition of σ^i. The first cycle will be:

$$(1 \;\; R_m^+(1+i) \;\; R_m^+(R_m^+(1+i)+i) \;\; \cdots)$$

It can be re-written:

$$(1 \;\; R_m^+(1+i) \;\; R_m^+(1+2i) \;\; \cdots \;\; R_m^+(1+(l-1)i))$$

Here l is the length of this cycle, and $\sigma^i(R_m^+(1+(l-1)i)) = 1$. This implies that $R_m^+(1+li) = 1$. Note that $1 \le l \le m$ and $1 \le k < l \implies R_m^+(1+ki) \neq 1$. This immediately implies that l is the least positive integer s.t. $1 + li = 1 + nm$. Let $d = \gcd(m, i)$. Then, $l = m/d$. So, $l = m \iff d = 1$. This completes the proof.

□

We can improve above result into nice theorem.

Corollary: i-th power of cycle

Let σ be the m-cycle. For $i \in \mathbb{Z}$, let $d = \gcd(m, i)$. Then, σ^i is decomposed into d of length (m/d)-cycles.

Exercise 13

Show that an element has order 2 in S_n if and only if its cycle decomposition is a product of commuting 2-cycles.

Proof. It can be easily obtained, since we already know that disjoint cycles commute, and for m-cycle σ, $|\sigma| = m$. □

Exercise 14

Let p be a prime. Show that an element has order p in S_n if and only if its cycle decomposition is product of commuting p-cycles. Show by an explicit example that this need not be the case if p is not prime.

Proof. ($\Rightarrow$) Suppose that there exists σ' in the cycle decomposition of σ whose order is p such that $\sigma' = (a_1 \ \cdots \ a_n)$ and $n \neq p$. Then here, since cycle decomposition is constructed by disjoint cycles,

$$\sigma^p(a_k) \mapsto a_{R_n^+(k+p)}$$

And since p is prime, this never be a_k. So that, σ^p is not identity permutation on a_k, contradicts that $|\sigma| = p$.
However, if p need not to be a prime, we can't guarantee that $R_n^+(k+p) \neq k$. It can be possible when p is multiple of n. Let's imagine

$$\sigma = (1\ 2)(3\ 4\ 5\ 6)$$

This have order 4. This completes the proof.

□

Proposition 1.3.1.

The order of $\sigma \in S_n$ equals the least common multiple of the lengths of the cycles in its cycle decomposition.

Proof. Let $\sigma = \sigma_1\sigma_2\cdots\sigma_k$ be the cycle decomposition of σ. Then since σ_i, σ_j commute, so

$$\sigma^i = \sigma_1^i\sigma_2^i\cdots\sigma_k^i$$

To this become identity permutation, one is clear that

$$\sigma_j^i = I \quad \forall 1 \leq j \leq k$$

Since we know $|\sigma_j| =$ length of σ_j denote by $l(j)$, $i = l(j)\cdot n_j, \quad \forall 1 \leq j \leq k$. To i be an order, i must be minimum positive one, so it is just definition of least common multiple of $l(j)$.

□

Here, I note that something I realized: For $\sigma \in S_m$, we can find the cycle decomposition of σ. There are length 1 upto m cycles. Let k_i be the number of i-cycles. Then, following holds:

$$m = \sum_{i=1}^{p} i \cdot k_i, \quad |\sigma| = \text{lcm}(i|k_i > 0)$$

And if the $\{k_i\}$ fixed, we can find the number of cases by

$$\frac{p!}{\sum_{i=1}^{p} i^{k_i} \cdot k_i!}$$

1.4 Matrix Groups

Actually, field is our later interests. However, to understand group theory, very powerful tool is understanding linear algebra. So we alert matrix groups, whose coefficients come from fields. For sake of goal, let's describe what the fields are.

Fields

A *field* is a set F with two **commutative** binary operations $+, \cdot$ on F such that $(F, +)$ forms an abelian group with identity 0, and $(F \setminus \{0\}, \cdot)$ also forms an abelian group with identity 1, and following *distributive law* holds:

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c), \forall a, b, c \in F$$

And we denote $F^{\times} = F \setminus \{0\}$

Actually our interests are groups of matrices over F. Let's define following important group here.

Matrix groups($GL_n(F)$)

$$GL_n(F) = \{A | A \in Mat_{n \times n}(F), \det(A) \neq 0, \text{i.e., invertible}\}$$

Let binary operation $\cdot$ denote matrix multiplication. It is easy to see that $(GL_n(F), \cdot)$ is a group. It called *general linear group of degree n*.

1.5 Quaternion Group

In mathematics, there are some famous counter-examples when we conject some properties on mathematical structures. This group is sort of them.

Quaternion Group

The *quaternion group* Q_8 is defined by

$$Q_8 := \{1, -1, i, -i, j, -j, k, -k\}$$

Here, we define binary operation $\cdot$ as follows;

$$\forall q \in Q_8, 1 \cdot q = q \cdot 1 = q$$

$$(-1) \cdot (-1) = 1, \quad \forall q \in Q_8, (-1) \cdot q = q \cdot (-1) = -q$$

$$i \cdot i = j \cdot j = k \cdot k = -1$$

$$i \cdot j = k \quad j \cdot i = -k$$

$$j \cdot k = i \quad k \cdot j = -i$$

$$k \cdot i = j \quad i \cdot k = -j$$

(I'll add more intuition about this group after I understand well.)

1.6 Homomorphisms and Isomorphisms

By defining groups, we abstracted many mathematical structures into group theories. Then now, we can imagine the maps between them and what the 'look the same group' is. First, let's define the structure-preserved maps on groups.

Homomorphism

Let $(G, \star), (H, \diamond)$ be groups. A map $\varphi : G \to H$ such that

$$\forall x, y \in G, \varphi(x \star y) = \varphi(x) \diamond \varphi(y)$$

is called *homomorphism* from G to H.

Then now, we define more strong (and more precise to our goal) concepts: Two groups are conceptually equal.

Isomorphism

Let the map $\varphi : G \to H$ be an homomorphism from G to H. We call that φ is an *isomorphism* when φ is a bijection. If such isomorphism exists, then we denote $G \cong H$ and say that G and H are *isomorphic*.

On the high level, $G \cong H$ if there exists bijective structure-preserve map φ. And if $G \cong H$, they are exactly same group except that how we write the elements and operation.

One of the most familar example is that $(\mathbb{R}, +) \cong (\mathbb{R}^+, \times)$. Such isomorphism is explicitly given as $\varphi(x) = e^x$. It is bijective since $\varphi^{-1}(a) = \log a$. And clearly homomorphism, since $e^{x+y} = e^x e^y$. So, both are exactly same structure.

Following fact is quite not obvious:

First isomorphim classes

Any non-abelian group of order 6 is isomorphic to S_3. Further, every group whose order is 6 is isomorphic to one of two groups: $S_3, \mathbb{Z}/6\mathbb{Z}$ and so, $S_3 \ncong \mathbb{Z}/6\mathbb{Z}$.

One of main goal in group theory is that learn skills to prove this kind of statements.

Now, I add some useful homomorphism criterion here. Suppose that G is generated by $\{s_1, s_2, \cdots, s_m\}$ and also has relations between them. Let H be the another group and $r_1, \cdots, r_m \in H$. If all the relations in G is satisfied in H when replace r_i into s_i, then there is a unique homomorphism $\varphi : G \to H$ which $\varphi(s_i) = r_i$. Further, if H is generated by $\{r_i\}$, then φ is surjective. Also, if $|G| = |H| < \infty$, then φ is injective. So in this case we can also determine $G \cong H$.

Here, I put some interesting Exercise problems that I proved.

Exercise Ch 1.6.

1. If $\varphi : G \to H$ is an isomorphism, prove that $|\varphi(x)| = |x|$ for all $x \in G$. Deduce that any two isomorphic groups have same number of elements of order n for each $n \in \mathbb{Z}^+$. Is the result true if φ is only assumed to be a homomorphism?

2. If $\varphi : G \to H$ is an isomorphism, prove that G is abelian if and only if H is abelian. If $\varphi : G \to H$ is a homomorphism, what additional conditions are sufficient to ensure that if G is abelian then so is H?

3. Prove that if $|\Delta| = |\Omega|$ then $S_\Delta \cong S_\Omega$

4. $\varphi : G \to H$ is homomorphism. Then $\varphi(G) < H$ and if φ is injective then $G \cong \varphi(G)$.

5. $\varphi : G \to H$ is homomorphism. Define $\ker(\varphi) = \{g \in G | \varphi(g) = 1\}$. Prove that $ker(\varphi) < G$ and $ker(\varphi) = \{1_G\}$ iff φ is injective.

6. Let G be any group. $\varphi : G \to G$ define by $g \mapsto g^{-1}$. Prove that φ is homomorphism iff G is Abelian.

7. Same question as above, by $g \mapsto g^2$.

8. A is abelian group and fix $k \in \mathbb{Z}$. $a \mapsto a^k$ is homomorphism from A to A.

9. Let G be a finite group which possesses an automorphism σ such that $\sigma(g) = g$ iff $g = 1$. If σ^2 is the identity map, then G is abelian. Hint: $g \mapsto x^{-1}\sigma(x)$.

10. G is finite group. $x, y \in G, |x| = |y| = 2$ and $G = \langle x, y\rangle$. For $n = |xy|$, $G \cong D_{2n}$.

1.7 Group Actions

Group actions are natural concepts, as we can imagine the $GL_n(\mathbb{R})$ as a linear transformation on $\mathbb{R}^n$ space. They 'act' on vector in $\mathbb{R}^n$, while preserves the group structures between such linear transformations. So we can define group action very naturally:

Group Action

A *(left) group action* of a group G on a set A is a map $G \times A \to A$ written as $g \cdot a$, for every $g \in G, a \in A$, such that

1. $g_1 \cdot (g_2 \cdot a) = (g_1 \cdot g_2) \cdot a, \forall g_1, g_2 \in G, a \in A$
2. $1 \cdot a = a, \forall a \in A$

Now, we'll make quite interesting observation. That's the relations between group actions and permutations.

Group Action as Permutation

For every $g \in G$, we can define a map

$$\sigma_g : A \to A$$

$$\sigma_g : a \mapsto g \cdot a$$

Then, $\sigma_g \in S_A$ and $\varphi_G : g \mapsto \sigma_g$ is a homomorphism of $G \to S_A$.

This $\varphi : G \to S_A$ is called permutation representation associated to the given action G on A. i.e., If we have G action on A, we can find homomorphism from G to S_A always. Another direction is also hold. If we have $\varphi : G \to S_A$ a homomorphism, then we can define G-action on A.

faithful kernel left regular action

We say that a G action on A is *faithful* if distinct elements of G induce distinct permutations of A. We say that the *kernel* of G action on A be $\{g \in G \mid g \cdot a = a, \quad \forall a \in A\}$ Let A be a set $A = G$. Define action of G on A by $g \cdot a = ga$. This action is called *left regular action* of G on itself.

Exercise Sec 1.7.

1. Prove that a group G acts faithfully on a set A if and only if the kernel of the action is $\{1_G\}$.
2. Assume n is an even positive integer and show that D_{2n} acts on the set consisting of pairs of opposite vertices of a regular n-gon. Find the kernel of this action.

3. Let G be any group and $A = G$. Show that $g \cdot a = gag^{-1}$ is group action of G on A.

4. So, each $g \in G$ define a map $x \mapsto gxg^{-1}$. Prove that this is an isomorphism from G to itself.

5. Let H be a group acting on a set A. Prove that the relation $\sim$ on A defined by

$$a \sim b \iff a = hb \text{ for some } h \in H$$

is an equivalence relation. These equivalence classes is called *orbit* of x under the action H.

6. If G is a finite group and $H < G$ then $|H|$ divides $|G|$. (Lagrange's Theorem)

7. Group of rigid motions of a tetrahedron is isomorphic to a subgroup of S_4.

Chapter 2

Subgroups

2.1 Definition and Examples

Subgroup

Let G be a group. $H \subseteq G$ is called *subgroup* of G if H is nonempty and H is closed under products and inverses of G. We denote $H \leq G$.

Every subgroup must contain 1_G. It can be the most easy assertation when $H \leq G$.

Subgroup Criterion

$H \leq G$ if and only if

- $H \neq \emptyset$, i.e. $1_G \in H$
- $\forall x, y \in H, xy^{-1} \in H$

If H is finite, then it is sufficient to show that H is nonempty and closed under multiplication.

Exercise Sec 2.1.

1. Give an explicit example of group G and an infinite subset H of G that is closed under the group operation but is not a subgroup of G.
2. Prove that G cannot have a subgroup H with $|H| = n-1$, where $n = |G| > 2$
3. Let G be an abelian group. Prove that $\{g \in G \mid |g| < \infty\}$ is a subgroup of G (called the *torsion subgroup* of G). Give an explicit example where this set is not a subgroup when G is non-abelian.
4. Let H and K be subgroups of G. Prove that $H \cup K$ is a subgroup of G if and only if either $H \subseteq K$ or $K \subseteq H$.

5. Let $G = GL_n(F)$, where F is any field. Define

$$SL_n(F) = \{A \in GL_n(F) \mid \det(A) = 1\}$$

(called the *special linear group*). Prove that $SL_n(F) \leq GL_n(F)$.

6. Prove that the intersection of an arbitrary nonempty collection of subgroups of G is again a subgroup of G. (Do not assume the collection is countable)

7. Let H be an subgroup of the additive group of rational numbers with the property that $1/x \in H$ for every nonzero element x of H. Prove that $H = 0$ or $\mathbb{Q}$

8. Let $H_1 \leq H_2 \leq \cdots$ be can ascending chain of subgroups of G. Prove that $\cup_{i=1}^{\infty} H_i$ is a subgroup of G.

2.2 Centralizers, Normalizers, Stabilizers, Kernels

Centralizer

Let G be a group and $A \subseteq G$. Define the *centralizer* of A in G by

$$C_G(A) := \{g \in G \mid \forall a \in A, a = gag^{-1}\}$$

In other words, $C_G(A)$ is subset of G such that commutes with every element of A.

Lemma 2.2.1. *For any $A \subseteq G$, $C_G(A) \leq G$*

Proof. Clearly $1_G \in C_G(A)$, so $C_G(A) \neq \emptyset$ for any A. Suppose that $x, y \in C_G(A)$. Then,

$$\begin{aligned}\forall a \in A, &(xy^{-1})a(xy^{-1})^{-1} \\ &= (xy^{-1})a(yx^{-1}) \\ &= x(y^{-1}ay)x^{-1} \\ &= a\end{aligned}$$

Hence, $C_G(A) \leq G$.

□

Center

Define the *center* of G by

$$Z(G) = C_G(G) = \{g \in G \mid \forall x \in G, gx = xg\}$$

So we can treat this as a special case of centralizer.

Normalizer

Let denote $gAg^{-1} = \{gag^{-1} \mid a \in A\}$. Define the *normalizer* of A in G by

$$N_G(A) = \{g \in G \mid gAg^{-1} = A\}$$

Note that, it does not need to exactly commute with each elements of A.

Lemma 2.2.2. *For any $A \subseteq G, N_G(A) \leq G$.*

Proof. Same to above. □

We defined centralizer and normalizer for the 'subsets' of G. We can generalize this into notion of G-action on 'any set'.

Stabilizer

Suppose that G is a group acting on set S. For fixed $s \in S$, we define *stabilizer* of s in G by

$$G_s = \{g \in G \mid g \cdot s = s\}$$

Lemma 2.2.3. *Let G be a group acting on set S. For $s \in S$, $G_s \leq G$.*

Proof. The axiom of group action gives us that $1_G \cdot s = s$, so $1_G \in G_s$. And for $g_1, g_2 \in G_s$,

$$(g_1 g_2^{-1}) \cdot s = g_1 \cdot (g_2^{-1} \cdot s) = g_1 \cdot (g_2^{-1} \cdot (g_2 \cdot s)) = g_1 \cdot s = s$$

So $g_1 g_2^{-1} \in G_s$, hence $G_s \leq G$.

□

Kernel

Let G be a group acting on set S. The *kernel* of left G-action on S is defined by

$$\{g \in G \mid \forall s \in S, g \cdot s = s\}$$

Connections between stabilizer and normalizer

Suppose that G acts on $\mathcal{P}(G)$, collection of all subsets of G. Then action is defined by:

$$G \times \mathcal{P}(G) \to \mathcal{P}(G)$$

By conjugation action:

$$g : S \mapsto \{gsg^{-1} \mid s \in S\}$$

Then stabilizer of S in G is normalizer of S in G.
Similarly, suppose that $N_G(A)$ acts on $A \subseteq G$ by conjugation:

$$n : a \mapsto nan^{-1}$$

Then, kernel of this action is center of A in G.

Exercise Sec 2.2.

1. Prove that $C_G(Z(G)) = G$ and deduce that $N_G(Z(G)) = G$.

2. Prove that if $A \subseteq B \subseteq G$ then $C_G(B) \leq C_G(A)$.

3. Show that if $H \leq G$ then $H \leq N_G(H)$. Give an example that this is not necessarilly true if H is not a subgroup of G.

4. Show that if $H \leq G$ then $H \leq C_G(H)$ if and only if H is abelian.

5. Let $n \in \mathbb{Z}$ with $n \geq 3$. $Z(D_{2n}) = 1$ if n is odd, and $Z(D_{2n}) = \{1, r^k\}$ if $n = 2k$.

6. Let $G = S_n$, fix an $i \in [n]$ and let $G_i = \{\sigma \in G \mid \sigma(i) = i\}$ (the stabilizer of i in G). Use group actions to prove that G_i is a subgroup of G. Find $|G_i|$.

7. For any subgroup H of G and any nonempty subset A of G define $N_H(A)$ to be the set $\{h \in H \mid hAh^{-1} = A\}$. Show that $N_H(A) = N_G(A) \cap H$ and

deduce that $N_H(A)$ is a subgroup of H.

8. Let H be a subgroup of order 2 in G. Show that $N_G(H) = C_G(H)$. Deduce that if $N_G(H) = G$ then $H \leq Z(G)$.

9. Prove that $Z(G) \leq N_G(A)$ for any subset A of G.

2.3 Cyclic groups and Cyclic Subgroups

Cyclic Group

A group H is *cyclic* if H can be generated by a single element, i.e.,

$$\exists x \in H, H = \{x^n \mid n \in \mathbb{Z}\}$$

We denote $H = \langle x \rangle$ and H is *generated* by x. Cyclic group is always Abelian.

Proposition

If $H = \langle x \rangle$, then $|H| = |x|$. More specifically,

- if $|H| = n < \infty$, then $x^n = 1$ and $1, x, \cdots, x^{n-1}$ are all distinct elements of H
- if $|H| = \infty$, then $x^n \neq 1$ for all $n \neq 0$ and $x^a \neq x^b$ for all $a \neq b$ in $\mathbb{Z}$.

Proof. Suppose that $|x| = n < \infty$. Then, $\{1, x, \cdots, x^{n-1}\} \subseteq H$ and they are distinct. However, for $x^k \in H$, $x^k = x^{np+q} = (x^n)^p \cdot x^q = x^q \in \{1, x, \cdots, x^{n-1}\}$. So, $H = \{1, x, \cdots, x^{n-1}\}$ and $|H| = n$. Suppose that $|x| = \infty$. Then, for $a \neq b$, if $x^a = x^b$ then $x^{b-a} = 1$ so order $|x| \leq |b - a| < \infty$ contradiction. So, $|H| = \infty$.

□

Proposition

Let G be an arbitrary group, $x \in G$ and $m, n \in \mathbb{Z}$. If $x^n = 1, x^m = 1$ then $x^d = 1$, where $d = (n, m)$. In particular, if $x^m = 1$ for some $m \in \mathbb{Z}$ then $|x|$ divides m

Proof. Euclidean algorithm gives us that $\exists r, s \in \mathbb{Z}, d = (n, m) = nr + ms$. Then,

$$x^d = x^{(n,m)} = x^{nr+ms} = x^{nr} \cdot x^{ms} = 1$$

Suppose that $|x| = n$. Then, $x^m = x^{np+q} = x^q = 1$ for $0 \leq q < n$. Since $|x| = n$, $q = 0$. So, $m = np$, implies that $|x|$ divides m.

□

Theorem

Any two cyclic groups of the same order are isomorphic. More specifically,

- if $n \in \mathbb{Z}^+$ and $\langle x \rangle, \langle y \rangle$ are both cyclic groups of order n, then

$$\varphi : \langle x \rangle \to \langle y \rangle$$
$$x^k \mapsto y^k$$

  is well defined and is an isomorphism.

- if $\langle x \rangle$ is an infinite cyclic group, the map

$$\varphi : \mathbb{Z} \to \langle x \rangle$$
$$k \mapsto x^k$$

  is well defined and is an isomorphism.

Proof. To show that φ is well-defined, sufficient to show that

$$\forall r, s \in \mathbb{Z}, \quad x^r = x^s \implies \varphi(x^r) = \varphi(x^s)$$

Here, $x^{r-s} = 1$ implies that for $|x| = |y| = n$, $n|(r - s)$. Then, $r = nk + s$ holds. So,

$$\varphi(x^r) = \varphi(x^{nk+s}) = y^{nk+s} = y^s = \varphi(x^s)$$

So φ is well-defined. Also, it is easy to show that φ is a homomorphism from $\langle x \rangle$ to $\langle y \rangle$. Finally, this map is surjective and both groups have same finite order, this is an isomorphism. For the second one, do similarly.

□

From here, we'll denote that Z_n be the order n cyclic group, since all of them are isomorphic. Also, we can think that $Z_n \cong \mathbb{Z}/n\mathbb{Z}$. However, a given cyclic group may have more than one generator.

Proposition

Let G be a group, $x \in G$ and let $a \in \mathbb{Z} - \{0\}$.

- If $|x| = \infty$, then $|x^a| = \infty$.
- If $|x| = n < \infty$, then $|x^a| = \frac{n}{(n,a)}$
- In particular, if $|x| = n < \infty$ and a is a positive integer dividing n, then $|x^a| = \frac{n}{a}$.

Proof. Only second one is meaningful to prove.

□

Proposition

Let $H = \langle x \rangle$.

- $|x| = \infty$ then $H = \langle x^a \rangle$ if and only if $a = \pm 1$
- $|x| = n < \infty$ then $H = \langle x^a \rangle$ if and only if $(a, n) = 1$.

Theorem

Let $H = \langle x \rangle$ be a cyclic group.

1. Every subgroup of H is cyclic. More precisely, if $K \leq H$ then either $K = \{1\}$ or $K = \langle x^d \rangle$ where d is smallest positive integer such that $x^d \in K$.
2. If $|H| = \infty$, then for any distinct nonnegative integers a, b, $\langle x^a \rangle \neq \langle x^b \rangle$. Futhermore, for every integer m, $\langle x^m \rangle = \langle x^{|m|} \rangle$, so that the nontrivial subgroups of H correspond bijectively with the integers $1, 2, 3, \cdots$.
3. If $|H| = n < \infty$, then for each positive integer a dividing n there is a unique subgroup of H of order a. This subgroup is cyclic group $\langle x^d \rangle$ where $d = n/a$. Futhermore, for every integer m, $\langle x^m \rangle = \langle x^{(n,m)} \rangle$ so that the subgroups of H correspond bijectively with the positive divisors of n.

Proof.

1. For such $x^d \in K$, $\langle x^d \rangle \leq K$ is obvious. However, if there exists $x^m \in K$ such that $x^m \notin \langle x^d \rangle$, then we can find $k \in \mathbb{Z}$ such that $0 < m - dk < d$, and since $x^{m-dk} \in K$, this is contradiction to choice of d.
2. Easy to show.
3. Suppose that K be any subgroup of order a. Then by (1), $K = \langle x^b \rangle$. Then here,

$$\frac{n}{d} = a = |K| = |x^b| = \frac{n}{(n, b)}$$

This implies that $d = (n, b)$. So $d|b$ and implies that $x^b \in \langle x^d \rangle$. Hence $\langle x^d \rangle \leq \langle x^b \rangle$. However they have same order, $K = \langle x^d \rangle$.

□

Exercise Sec 2.3.

Chapter 3

Quotient Groups, Homomorphisms

3.1 Definitions

The ultimate goal of group theory is understanding the structure of groups, and then classify them. For that, we need to understand 'smaller' groups contained in original group. We discuss about subgroups in previous section. Actually, there are another meaningful smaller groups, namely *quotient groups*.

Fibers and Homomorphism

Suppose that $\varphi : G \to H$ is a group homomorphism. The *fibers* of φ are the sets of elements of G projecting to single elements of H. Actually, H can be intuitively 'larger' group than G. However, the image of G, namely $\varphi(G)$ is at most large as G. So, a *fiber* is elements of G such that zipped into one points of $\varphi(G)$. For any $a \in \varphi(G) \subseteq H$, we can define $X_a \subseteq G, \forall x \in X_a, \varphi(x) = a$ as a *fiber above a*.

We'll dive into these interesting subgroups of G during this chapter.

Kernel

Let $\varphi : G \to H$ be a homomorphism. The *kernel* of φ is

$$\ker \varphi := \{g \in G \mid \varphi(g) = 1_H\}$$

i.e., *kernel* is fiber above 1_H.

Proposition 1

Let G, H be groups and $\varphi : G \to H$ be a homomorphism.

1. $\varphi(1_G) = 1_H$
2. $\varphi(g^{-1}) = \varphi(g)^{-1}$, for all $g \in G$
3. $\varphi(g^n) = \varphi(g)^n$, for all $g \in G$
4. $\ker \varphi \leq G$
5. $\varphi(G) \leq H$

Quotient Groups

Let $\varphi : G \to H$ be a homomorphism with $\ker \varphi = K$. The *quotient group*, G/K is the group whose elements are the fibers of φ with the group operation : $X_a \cdot X_b = X_{ab}$

In this definition, we intuitively know how they are working. But one is difficult to formalize these things and use as a tool to understand the group structure of G. So, we'll view quotient group and homomorphisms as alternating perspective.

Proposition 2

Let $\varphi : G \to H$ be a homomorphism and $\ker \varphi = K$. For a fiber on a, $\varphi^{-1}(a), X_a \in G/K$,

$$\forall u \in X_a, X_a = \{uk \mid k \in K\} = \{ku \mid k \in K\}$$

Means that, we can enumerate all elements of any fibers by **one element** and **kernel**.

Cosets

For any $N \leq G$ and $g \in G$,

$$gN = \{g \cdot n \mid n \in N\}, \quad Ng = \{n \cdot g \mid n \in N\}$$

called respectively a *left coset* and *right coset* of N in G. Any element of a coset is called a *representative* for the coset.

Proposition 3.1.1. *For homomorphism $\varphi : G \to H$ with $K = \ker \varphi$, every fiber is a left(right) coset of K in G. And, if $N \leq G$ is kernel of some homomorphism, $gN = Ng$ for any $g \in G$. To represent coset, the choice of element does not matter. Means that, $gN = g'N$ if g, g' are in same coset.*

This proposition gives us that, when we discuss quotient groups, it can be viewed as a set of left cosets.

Theorem 3

Let G be a group and K is the kernel of some(unknown) homomorphism from G. Then, collection of every left cosets of K in G:

$$\{gK \mid \forall g \in G\}$$

with following binary operation:

$$uK \cdot vK = (uv)K$$

forms group, G/K. Sometimes we denote $\bar{u}$ to coset uK, the coset of K containing u.

This theorem is quite nice since we can understand G/K without explicit homomorphism and computation of fibers **if K is kernel of some homomorphism**. Then natural question is that, what can we talk about **kernel of some homomorphism**?

Proposition 4

Let $N \leq G$. The set of left cosets of N in G forms a partition of G. Furthermore, $\forall u, v \in G, uN = vN$ if and only if $v^{-1}u \in N$ if and only if u, v are in the same coset.

Proof. For any $g \in G$, there exists coset $gN \subseteq G$, so

$$\bigcup_{g \in G} gN = G$$

Now it is enough to show that, all distinct cosets are disjoint. Suppose that $g_1N \neq g_2N$. If $\exists x \in g_1N \cap g_2N$, $x = g_1n_1 = g_2n_2$ by definition. Then, $g_1 = g_2n_2n_1^{-1} = g_2m$, here $m \in N$ since $N \leq G$ so $g_1N \subseteq g_2N$ and vice versa, $g_2N \subseteq g_1N$. This is contradiction, so $g_1N \cap g_2N = \emptyset$.

□

Proposition 5

Let $N \leq G$.

1. The operation on set of left cosets of N in G described by

$$uN \cdot vN = (uv)N$$

   is well-defined if and only if $gng^{-1} \in N$ for all $g \in G$ and all $n \in N$. i.e., $N_G(N) = G$.

2. If the above operation is well-defined, then it makes group G/N.

Proof. ($\Rightarrow$) The well-definedness implies that

$$\forall u, u_1 \in uN, \forall v, v_1 \in vN, \quad uvN = u_1v_1N$$

Then, $n, 1 \in N, g^{-1} \in g^{-1}N$, we can obtain from proposition 4:

$$ng^{-1}N = g^{-1}N \iff gng^{-1} \in N$$

($\Leftarrow$) Suppose that $N_G(N) = G$. Then, $\forall u, u_1 \in uN, u_1 = un$ and similarly, $v_1 = vm$. Then,

$$u_1v_1 = (un)(vm) = u(vv^{-1})nvm = uv(v^{-1}nv)m = uvm' \in uvN$$

□

Normal Subgroup

For $N \leq G$, if $N_G(N) = \{g \in G \mid gng^{-1} \in N\} = G$ then N is *normal subgroup* of G and denote $N \trianglelefteq G$

Proposition 3.1.2. *G/N is subgroup of G if and only if $N \trianglelefteq G$ if and only if $gN = Ng$ for all $g \in G$.*

Proposition 3.1.3. *To determine that $N \trianglelefteq G$, if we know $N = \langle A \rangle$, it is enough to show that $gag^{-1} \in N$ for any $a \in A, g \in G$. Further, if we know $G = \langle B \rangle$, it is sufficient to show that $bnb^{-1} \in N$ for any $n \in N, b \in B$.*

Proposition 7

$N \trianglelefteq G \iff N = \ker\varphi \iff G/N$ is group.

Definition 3.1.4. Here, we can easily find above $\varphi : G \to G/N$ by $g \mapsto gN$. This is obviously homomorphism whose kernel is N, is called *natural homomorphism* of G onto G/N.

!Caution!

Note that $A \trianglelefteq B$, $B \trianglelefteq C$ does not implies that $A \trianglelefteq C$. i.e., $\trianglelefteq$ does not transitive. The *normality* is relative.

Exercise Sec 3.1.

3.2 Cosets and Lagrange's Theorem

In this section, we'll analyze the subgroup structure of G. One of the most important feature of group is its order. One important theorem is following:

Lagrange's Theorem

G is finite group and $H \leq G$. Then, $|H|$ divide $|G|$ and the number of left cosets of H in G is $\frac{|G|}{|H|}$.

Proof. It is enough to show that the order of every coset is same. Then, since $1H = H$ also a coset and cosets form a partition of G, claim holds. Then we'll find bijection:

$$\forall g \in G, \quad H \to gH$$

This is injective, since if $gh_1 = gh_2$ then $h_1 = h_2$ by cancellation in G. Also surjective, obvious from the definition of gH. So clearly this is bijection, so

$$\forall g \in G, \quad |gH| = |H|$$

This completes the proof.

□

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Let G be a group (possibly infinite) and $H \leq G$, the number of left cosets of H in G is called *index* of H in G, and denoted by $|G : H|$

Corollary 9

Let G be a finite group and $x \in G$, then $|x|$ divides $|G|$, so $x^{|G|} = 1$

Proof. Since $x \in G$, we can construct $H = \langle x \rangle$ and $H \leq G$. Since G is finite, $|H|$ divide $|G|$. Since H is cyclic group, $|H| = |x|$, so $|x|$ divides $|G|$.

□

Corollary 10

Let G be a finite group and $|G| = p$ is prime, then $G \cong Z_p$

Proof. By above, for $x \in G$, $|x| = 1$ or p since p is prime. If $|x| = |G| = p$, then

$$G = \{1, x, \cdots, x^{p-1}\}$$

Then G is cyclic group $(= \langle x \rangle)$, so $G \cong Z_p$.

□

Example

Let G be any group and $H \leq G$, $|G : H| = 2$. Then, $H \trianglelefteq G$

Proof. Since $|G : H| = 2$, there are 2 cosets of H exists: H and $G - H = gH$. Suppose that there exists $h \in H$ such that $ghg^{-1} \in gH$. Then it implies that

$$\begin{aligned} ghg^{-1} = gh' &\implies hg^{-1} = h' \in H \\ &\implies g = h'^{-1}h \in H \end{aligned}$$

However, $g \in gH = G - H$. This is contradiction, so $\forall g \in G - H$ and $\forall h \in H$, $ghg^{-1} \in H$. For $g \in H$, it is obvious. So, $N_G(H) = G$ and $H \trianglelefteq G$.

□

Remarks

- 'Normal' does not means common. Normal subgroup is quite rare structure when G is given.
- Moreover, finding normal subgroup is hard problem.
- If G is finite group and n divides $|G|$, then G need not have a subgroup of order n.
- If G is abelian and n divides $|G|$, then subgroup whose order is n must exists.

Cauchy's Theorem

If G is a finite group and p is a prime dividing $|G|$ then G has an element of order p.

Proof. □

Sylow's Theorem

If G is a finite group of order p^α, where p is a prime and p does not divide m, then G has a subgroup of order p^α.

Now, we introduce another very natural construction.

Subgroup Multiplication

Let $H, K \leq G$. We define

$$HK = \{hk \mid h \in H, k \in K\}$$

Proposition 13

If H, K are finite subgroup then

$$|HK| = \frac{|H||K|}{|H \cap K|}$$

Proof. We can denote

$$HK = \bigcup_{h \in H} hK$$

We know that each cosets have same order $|K|$. So it is enough to show that there are $\frac{|H|}{|H \cap K|}$ distinct cosets. We already show that

$$\begin{aligned} h_1K = h_2K &\iff h_2^{-1}h_1 \in K \\ &\iff h_2^{-1}h_1 \in K \cap H \\ &\iff h_1(H \cap K) = h_2(H \cap K) \end{aligned}$$

Then, since $H, K \leq G$ implies that $H \cap K \leq H$, so number of distinct cosets forms hK is same with $|H : H \cap K| = \frac{|H|}{|H \cap K|}$.

□

!Caution!

We can't guarantee HK be a subgroup in general. See following proposition.

Proposition 3.2.1. $H, K \leq G$, $HK \leq G \iff HK = KH$

Proof. ($\Rightarrow$) If $HK \leq G$, then for any $k \in K, h \in H$, obtain $k, h \in HK$ so $kh \in HK$. So, $KH \subseteq HK$. Conversely, let $hk \in HK$. Then $k^{-1}h^{-1} = h'k' \in HK$.

$$hk = (h'k')^{-1} = k'^{-1}h'^{-1} \in KH$$

So, $HK \subseteq KH$.

($\Leftarrow$) If $HK = KH$, then for any $hk, h'k' \in HK$,

$$\begin{aligned} (hk)(h'k')^{-1} &= h(kk'^{-1})h'^{-1} \\ &= h(k_0h_0) \\ &= h(h_1k_1) \\ &= (hh_1)k_1 \in HK \end{aligned}$$

□

Corollary 15(For Second Isomorphism Theorem)

If $H, K \leq G$ and $H \leq N_G(K)$ then $HK \leq G$. Further, if $K \trianglelefteq G$ then $HK \leq G$ for any $H \leq G$.

Proof. Since $H \leq N_G(K)$, we know that $hKh^{-1} = K$ for any $h \in H$. Then, this directly implies that $hK = Kh$, so $HK = KH$. So, $HK \leq G$. If $K \trianglelefteq G$, then for any $H \leq G$, $H \leq N_G(K)$ obtained. This completes the proof. $\square$

Exercise Sec 3.2.

3.3 The Isomorphism Theorems

First Isomorphism Theorem

If $\varphi : G \to H$ is a homomorphism, then $\ker\varphi \trianglelefteq G$ and $G/\ker\varphi \cong \varphi(G)$

$$\begin{tikzcd}
G \arrow[rr, "\varphi\text{: Hom}"] \arrow[d, two heads] & & H \\
G/\ker\varphi \arrow[rr, leftrightarrow, "\cong", "\exists!\bar{\varphi}"'] \arrow[urr, phantom, "\circlearrowleft"] & & \varphi(G) \arrow[u, hook]
\end{tikzcd}$$

Corollary 17

Let $\varphi : G \to H$ be a homomorphism.

1. φ is injective if and only if $\ker\varphi = 1$
2. $|G : \ker\varphi| = |\varphi(G)|$

Proposition 3.3.1. *$\varphi : V \to W$ is a linear transformation, then* $\dim V = \text{rank } \varphi + \text{nullity } \varphi$

Proof.

$$\varphi(V) \cong V/\ker\varphi \implies \text{rank } \varphi = \dim V - \text{nullity}\varphi$$

$\square$

Second Isomorphism Theorem

If $A, B \leq G$ and $A \subseteq N_G(B)$ then $AB \leq G$ and $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$, and

$$(AB/B) \cong (A/(A\cap B))$$

$$\begin{tikzcd}
A \arrow[rr] \arrow[d, two heads] & & AB \arrow[d, two heads] \\
A/(A\cap B) \arrow[rr, leftrightarrow, "\cong", "\exists!"'] \arrow[urr, phantom, "\circlearrowleft"] & & AB/B
\end{tikzcd}$$

$$\begin{tikzcd}
& G \arrow[d, dotted, no head] & \\
& AB \arrow[dl] \arrow[dr] & \\
A \arrow[dr] & & B \arrow[dl] \\
& A\cap B \arrow[d, dotted, no head] & \\
& 1 &
\end{tikzcd}$$

Proof. As corollary 15 mentioned, we can obtain $AB \leq G$ directly. Next, since $A \subseteq N_G(B)$ and $B \subseteq N_G(B)$ is obvious, we can obtain that $AB \subseteq N_G(B)$ and so $B \trianglelefteq AB$.

This implies that AB/B is group and $a_1B \cdot a_2B = (a_1a_2)B$ is well-defined. Then now, let's imagine following mapping:

$$\varphi : A \to AB/B$$

work as

$$\varphi : a \mapsto (a \cdot 1)B = aB$$

Here, φ is clearly a homomorphism on A. The kernel of this map is

$$\ker \varphi = \{a \in A \mid aB = B\} = \{a \in A \mid a \in B\} = A \cap B$$

And φ is surjective, so $\varphi(A) = AB/B$. So we can apply First Isomorphism Theorem:

$$\begin{tikzcd}
A \arrow[r, two heads, "\varphi"] \arrow[d, two heads] & AB/B \\
A/(A \cap B) \arrow[r, leftrightarrow, "\cong"] & AB/B \arrow[u, hook]
\end{tikzcd}$$

So,

$$AB/B \cong A/(A \cap B)$$

□

Remark

More often, instead of $A \subseteq N_G(B)$ or $A \leq N_G(B)$, following condition also works.

$$B \trianglelefteq G$$

(Since this directly implies that $A \leq N_G(B)$)

Third Isomorphism Theorem

Let G be a group and $H \trianglelefteq G$, $K \trianglelefteq G$, $H \leq K$. Then $K/H \trianglelefteq G/H$ and

$$(G/H)/(K/H) \cong G/K$$

$$\begin{tikzcd}
G \arrow[r, two heads] \arrow[d, two heads] \arrow[dr, phantom, "\circlearrowleft"] & G/H \arrow[d, two heads] \\
G/K \arrow[r, dash, "\cong"] & (G/H)/(K/H)
\end{tikzcd}$$

Proof. For any $kH \in K/H$ and $gH \in G/H$, since $kH \in G/H$ and $H \trianglelefteq G$, following is well-defined.

$$(gH)(kH)(gH)^{-1} = (gH)(kH)(g^{-1}H) = (gkg^{-1})H \in K/H$$

So,

$$K/H \trianglelefteq G/H$$

Then, we can imagine following mapping:

$$\varphi : G/H \to G/K$$

works by

$$\varphi : gH \mapsto gK$$

Then φ is homomorphism and

$$\begin{aligned}\ker \varphi &= \{gH \in G/H \mid gK = K\} \\ &= \{gH \in G/H \mid g \in K\} \\ &= K/H\end{aligned}$$

So, we can draw First Isomorphism Theorem here:

$$\begin{tikzcd}
G/H \arrow[r, "\varphi"] \arrow[d, twoheadrightarrow] & G/K \\
(G/H)(K/H) \arrow[r, leftrightarrow, red, "\cong"] & G/K \arrow[u, hookrightarrow]
\end{tikzcd}$$

This shows that

$$(G/H)/(K/H) \cong G/K$$

□

Now, we want to use G/N as tool of understanding structure of G. Then one natural consequence is about their lattices. We'll observe following in Fourth Isomorphism Theorem.

$$\begin{tikzcd}
G \arrow[d, dash, dotted, "\text{Lattice of } G/N"] \\
N \arrow[d, dash, dotted, "\text{Lattice of } N"'] \\
1
\end{tikzcd}$$

Fourth Isomorphism Theorem

Let $N \trianglelefteq G$ then there exists a bijection $A \to \bar{A}$ between

$$\{A \leq G \mid N \subseteq A\} \leftrightarrow \{\bar{A} = A/N \leq G/N\}$$

And for $N \leq A, B \leq G$,

1. $A \leq B$ if and only if $\overline{A} \leq \overline{B}$
2. $A \leq B$ then $|B : A| = |\overline{B} : \overline{A}|$
3. $\overline{\langle A, B\rangle} = \langle \overline{A}, \overline{B}\rangle$
4. $\overline{A \cap B} = \overline{A} \cap \overline{B}$
5. $A \trianglelefteq G$ if and only if $\overline{A} \trianglelefteq \overline{G}$

Proof. For any $\overline{A} \leq G/N$, $\overline{A}$ is collection of cosets and coset binary operation is well-defined. So,

$$\forall a_1N, a_2N \in \overline{A}, \quad (a_1a_2)N \in \overline{A}$$

So, the complete preimage of $\overline{A}$ is disjoint union of each complete preimage of $aN \in \overline{A}$, noted by

$$A = \biguplus_{aN \in \overline{A}} A_a$$

Then, $\forall a, b \in A$, we know that $bN \in \overline{A}$, so $b^{-1}N \in \overline{A}$ and so $b^{-1} \in A$. So, we can obtain

$$ab^{-1} \in A \implies A \leq G$$

So, above bijection is surjective and $\overline{A} = A/N$.

Now suppose that $\overline{A_1} = \overline{A_2}$. Then as we've seen, $A_1/N = A_2/N$. It is not hard to see that $A_1 = A_2$.

□

Remarks

Let $\varphi : G \to G/N$ be natural homomorphism. Then, for $H \leq G$,

$$\varphi(H) = \varphi(HN)$$

Exercise Sec 3.3.

3.4

Chapter 4

Group Actions

4.1

4.2 Group Actions: Left Multiplication

4.3 Group Actions: Conjugation

4.4 Automorphisms

4.5 Sylow's Theorem

Chapter 5

Direct and Semidirect Products

5.1 Direct Products